

Michael Harhay

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EDUCATION

University of Toronto

Toronto, ON

Bachelor of Applied Science in Computer Engineering, Minor in AI Engineering Sept. 2022 – May 2027

Courses: Operating Systems, Algorithms and Data Structures, Computer Architecture, Compilers and Interpreters

Achievements: Dean's List Scholar (Fall 2024), achieved a 3.7 or higher in all programming & software courses thus far

SKILLS

Languages: C, C++, ASM, Verilog, Python, SQL, MATLAB, PowerShell, VBA, JavaScript, HTML/CSS

Systems & Hardware: Linux, Windows, FPGAs, PLC, CAN

Libraries & APIs: libc, POSIX, GTK, PyTorch, NumPy, pandas, Matplotlib, three.js

Tools: Git, GitHub, Docker, CMake, GCC, Clang, ModelSim, Logisim, VS Code, MS Office

Soft Skills: Problem Solving, Written & Verbal Communication, Leadership, Eagerness to Learn

EXPERIENCE

Software Engineering Co-op

May 2025 – August 2026

Arxtron Technologies

Mississauga, ON

- Developed industrial automation test software in C, interfacing with test equipment and DUTs.
- Initiated internal research and development projects, including a DevOps pipeline that streamlined software development and code quality checks using PowerShell, the GitHub API, and the Clang ecosystem.
- Conducted final inspection meetings with major automotive clients, troubleshooting issues and supporting successful project delivery.
- Worked with standard engineering documents such as project requirements, electrical schematics, and bills of materials to support project planning and execution.

Backend Software Lead

August 2024 – August 2025

U of T Creative Computing Association (UTCCA)

Toronto, ON

- Led the Backend Software team alongside two other executives, coordinating software development and project execution.
- Implemented a procedural terrain generation system using Perlin noise and three.js
- Designed an ambient music generation model utilizing a GAN framework. Implemented the model's data preprocessing and transformation pipeline with PyTorch and NumPy.

Intern, Branch Banking Performance Analysis

May 2024 – August 2024

TD Bank

Toronto, ON

- Developed a VBA macro to automate monthly executive reporting and binder generation.
- Cleaned and organized large datasets in Excel to support branch performance analysis.

PROJECTS

2048 Game (ECE243 Project) | C, FPGA, ASM, Git, GitHub

Jan. 2024 – May 2024

- Implemented the game 2048 in C for a NIOS-II processor on a DE1-SoC FPGA board.
- Developed framebuffer-based VGA graphics with double buffering and VSync for smooth animated tile movement.
- Implemented memory-mapped I/O to interface with VGA, PS/2 keyboard, and speakers.
- Spearheaded implementation and project organization within a two-person team, coordinating development tasks, deadlines, and version control.

EduRoute (ECE297 Project) | C++, GTK, Git, GitHub

Jan. 2024 – May 2024

- Designed and implemented a C++ mapping application using GTK and a provided OpenStreetMap API.
- Implemented a pathfinding mechanism, utilizing Dijkstra's algorithm before upgrading to an A* heuristic.
- Delivered a final presentation alongside teammates, pitching the application to professors.

INTERESTS

Music production & sound design, bass guitar, cooking and hosting, grand strategy games (Europa Universalis)